



**FACULTY OF ENGINEERING & TECHNOLOGY**

**PARUL INSTITUTE OF ENGINEERING &  
TECHNOLOGY**

**BACHELOR OF TECHNOLOGY**

**COMPUTER ORGANIZATION AND  
MICROPROCESSOR (303105211)**

**LABORATORY MANUAL**

# INDEX

| Sr. No.   | Aim                                                                                                                                                                                                                                                | Performance Date | Assessment Date | Sign | Marks |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|------|-------|
| <b>1</b>  | <b>PART A:</b> ADDITION OF TWO 8 BIT NUMBERS USING 8085.<br><b>PART B:</b> WRITE A PROGRAM TO ADD TWO 16-BIT NUMBERS STORED IN REGISTERS OR MEMORY LOCATIONS.<br><b>PART C:</b> 8 BIT SUBTRACTION                                                  |                  |                 |      |       |
| <b>2</b>  | <b>PART A:</b> WRITE AN 8085 ASSEMBLY LANGUAGE TO PERFORM MULTIPLICATION OF TWO 8 BIT NOS.<br><b>PART B:</b> WRITE AN 8085 ASSEMBLY LANGUAGE TO PERFORM DIVISION OF TWO 8 BIT NOS.                                                                 |                  |                 |      |       |
| <b>3</b>  | WRITE A PROGRAM TO ADD BLOCK OF 8-BIT DATA STORED IN MEMORY LOCATIONS.                                                                                                                                                                             |                  |                 |      |       |
| <b>4</b>  | <b>PART A:</b> WRITE AN 8085 ASSEMBLY LANGUAGE PROGRAM TO FIND THE MINIMUM FROM TWO 8-BIT NUMBERS.<br><b>PART B:</b> WRITE AN 8085 ASSEMBLY LANGUAGE PROGRAM TO GET THE MINIMUM FROM BLOCK OF N 8-BIT NUMBERS.                                     |                  |                 |      |       |
| <b>5</b>  | <b>PART A:</b> WRITE AN 8085 ASSEMBLY LANGUAGE PROGRAM TO FIND THE MAXIMUM FROM TWO 8-BIT NUMBERS.<br><b>PART B:</b> WRITE AN 8085 ASSEMBLY LANGUAGE PROGRAM TO GET THE MAXIMUM FROM BLOCK OF N 8-BIT NUMBERS.                                     |                  |                 |      |       |
| <b>6</b>  | <b>PART A:</b> WRITE AN ASSEMBLY LANGUAGE PROGRAM TO SORT DATA IN ASCENDING ORDER.<br><b>PART B:</b> WRITE AN ASSEMBLY LANGUAGE PROGRAM TO SORT DATA IN DECENDING ORDER.                                                                           |                  |                 |      |       |
| <b>7</b>  | <b>PART A:</b> WRITE AN 8085 ASSEMBLY LANGUAGE PROGRAM TO CONVERT GIVEN BCD NUMBER INTO ITS EQUIVALENT BINARY NUMBER.<br><b>PART B:</b> WRITE AN 8085 ASSEMBLY LANGUAGE PROGRAM TO CONVERT GIVEN BINARY NUMBER INTO ITS EQUIVALENT BCD NUMBER.     |                  |                 |      |       |
| <b>8</b>  | <b>PART A:</b> WRITE AN 8085 ASSEMBLY LANGUAGE PROGRAM TO CONVERT GIVEN BINARY NUMBER INTO ITS EQUIVALENT ASCII NUMBER.<br><b>PART B:</b> WRITE AN 8085 ASSEMBLY LANGUAGE PROGRAM TO CONVERT GIVEN ASCII NUMBER INTO ITS EQUIVALENT BINARY NUMBER. |                  |                 |      |       |
| <b>9</b>  | WRITE AN ASSEMBLY LANGUAGE PROGRAM IN 8085 CALCULATE THE SUM OF A SERIES OF EVEN NUMBERS.                                                                                                                                                          |                  |                 |      |       |
| <b>10</b> | WRITE AN ASSEMBLY LANGUAGE PROGRAM IN 8085 CALCULATE THE SUM OF SERIES OF ODD NUMBERS                                                                                                                                                              |                  |                 |      |       |